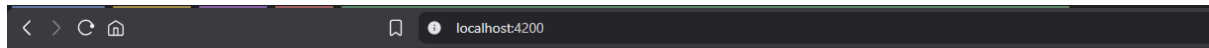


Angular Hands-on

HANDS-ON 1 – Environment Setup, Project Structure & First Component



Student Course Portal

[Home](#)[CoursesProfile](#)

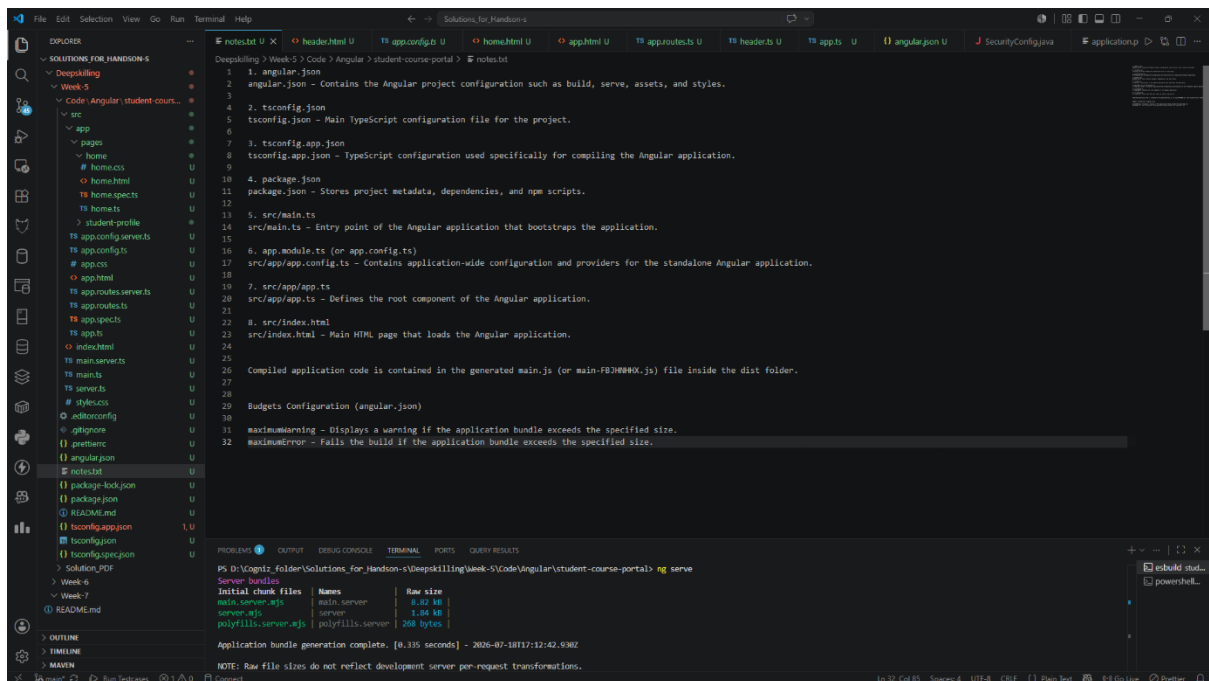
Welcome to Student Course Portal

This portal helps students browse courses, manage enrollment, and view their academic profile.

Courses Available: 12

Enrolled: 3

GPA: 3.8



```
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-5\Code\Angular\student-course-portal> ng serve
```

Browser bundles

Initial chunk files	Names	Raw size
main.js	main	7.54 kB
styles.css	styles	95 bytes
Initial total		7.64 kB

Server bundles

Initial chunk files	Names	Raw size
main.server.mjs	main.server	8.82 kB
server.mjs	server	1.84 kB
polyfills.server.mjs	polyfills.server	268 bytes

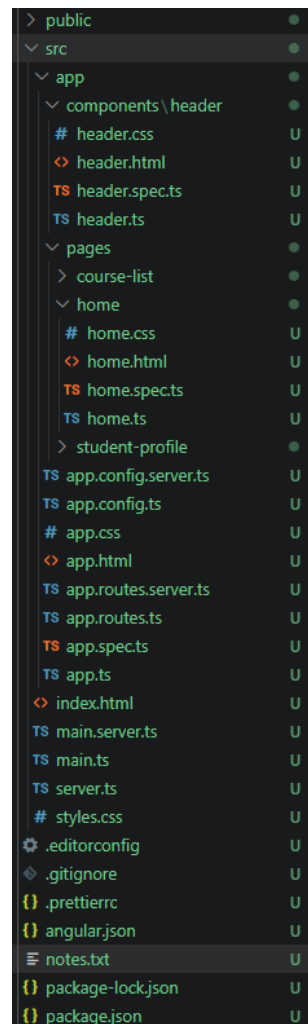
Application bundle generation complete. [0.335 seconds] - 2026-07-18T17:12:42.938Z

NOTE: Raw file sizes do not reflect development server per-request transformations.

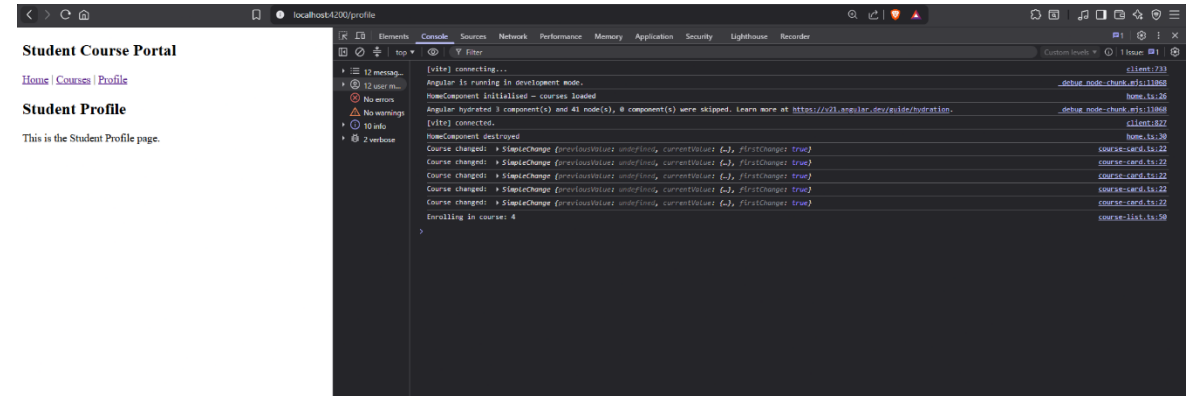
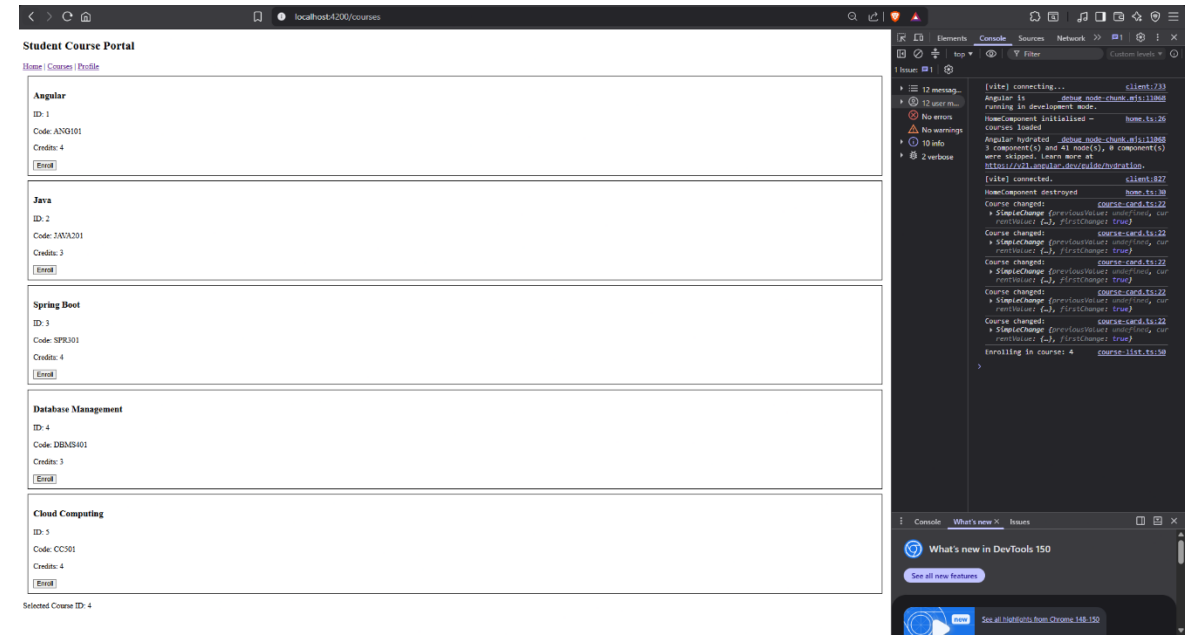
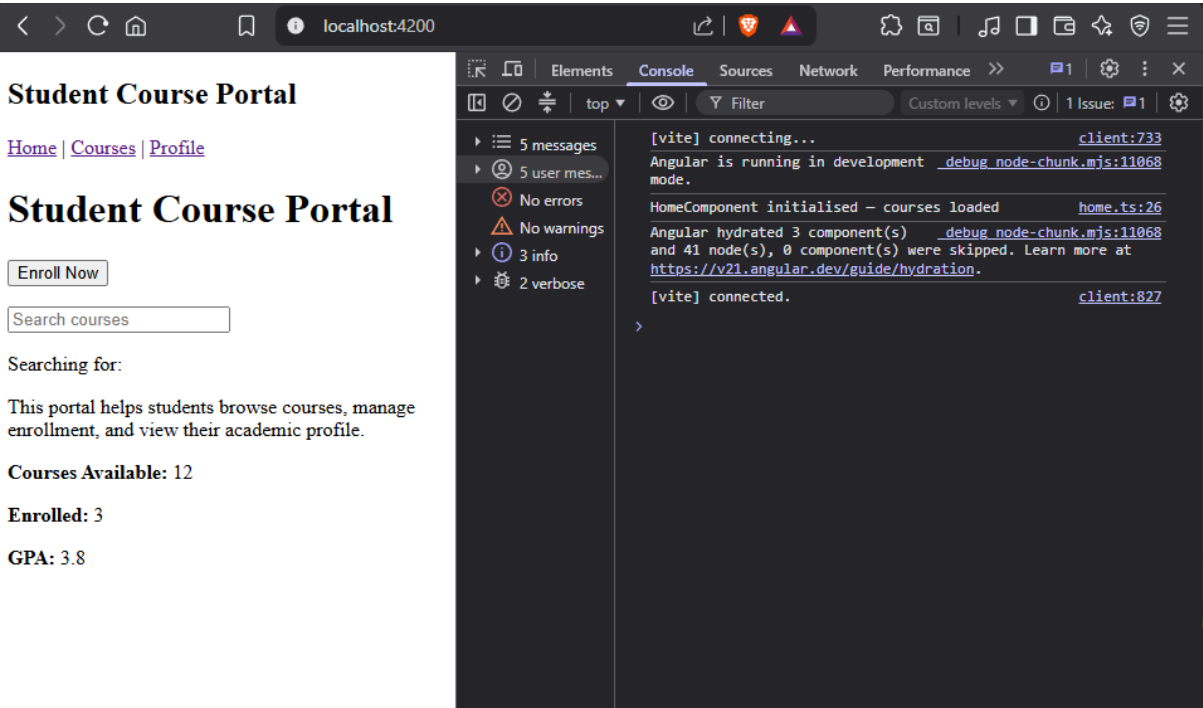
→ Local: <http://localhost:4200/>

→ press **h** + **enter** to show help

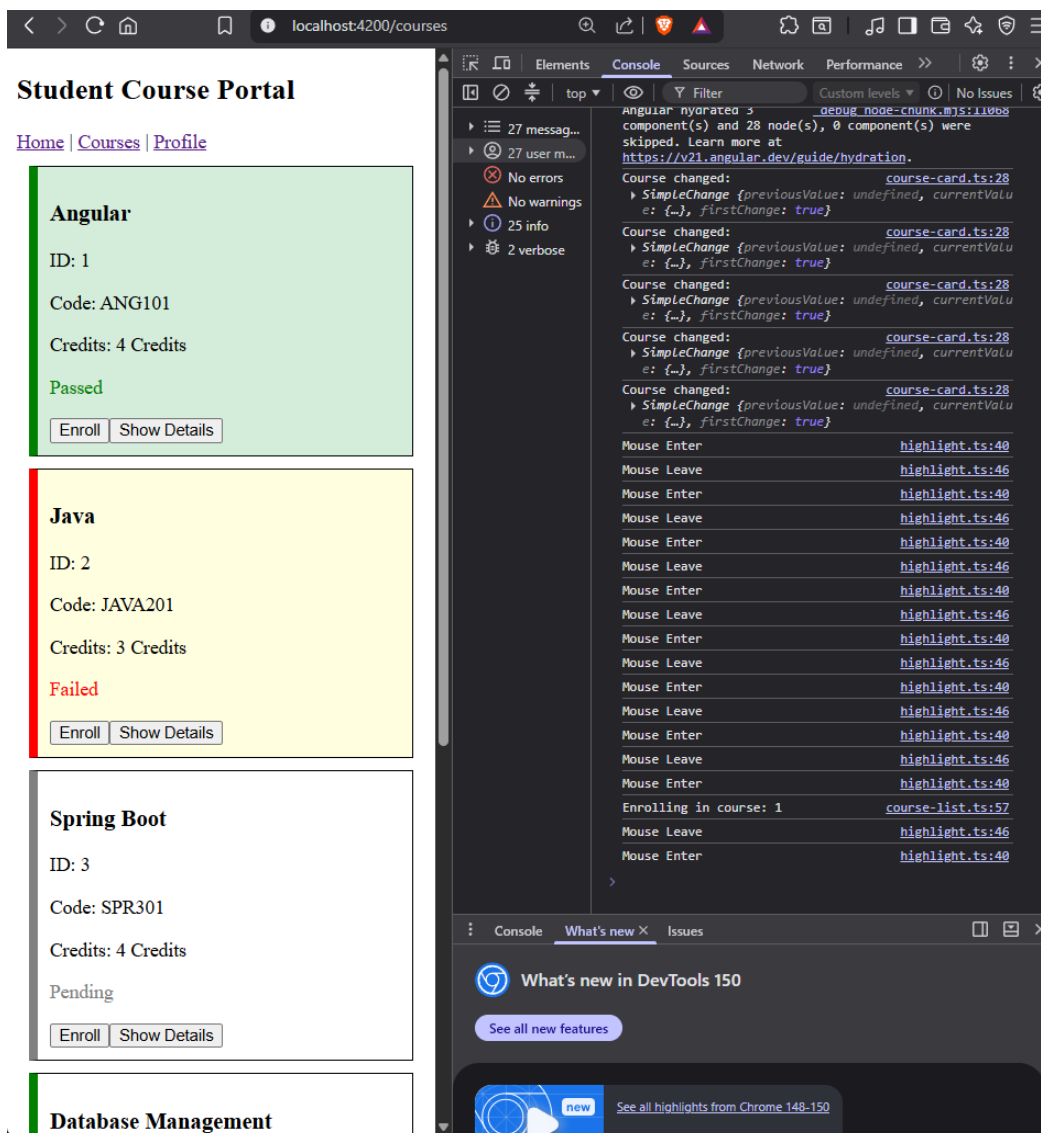
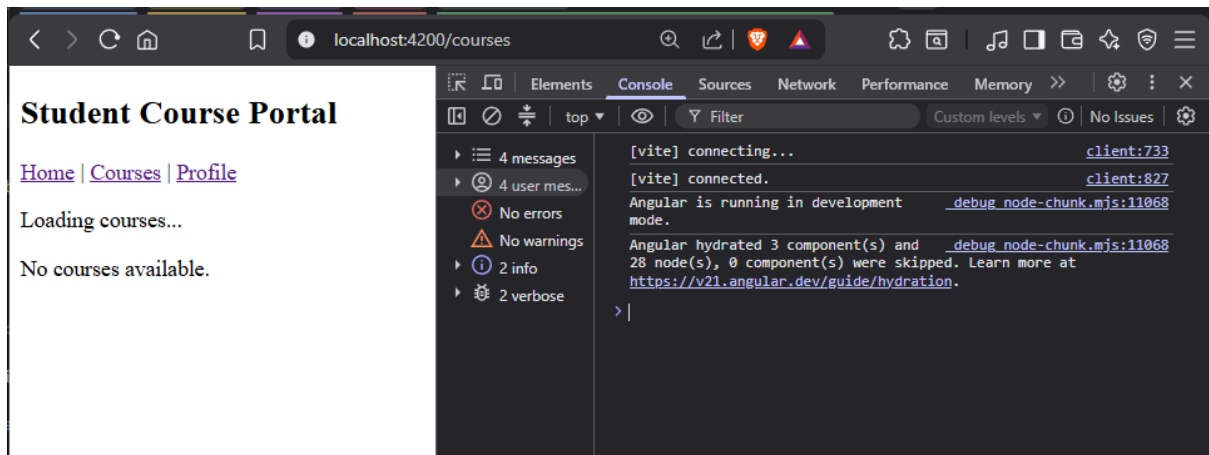
█



HANDS-ON 2 – Data Binding, Lifecycle Hooks & Component



HANDS-ON 3 – Directives & Pipes — Built-in and Custom



Spring Boot ID: 3 Code: SPR301 Credits: 4 Credits Pending Enroll Show Details
Database Management ID: 4 Code: DBMS401 Credits: 3 Credits Passed Enroll Show Details
Cloud Computing ID: 5 Code: CC501 Credits: 4 Credits Failed Enroll Show Details

Selected Course ID: 5

HANDS-ON 4 – Template-Driven Forms & Validation

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#)

Student Enrollment Form

Student Name	Utkarsh
Email	uckalinkar@gmail.com
Course ID	1
Preferred Semester	Odd v
☒ I Agree to the Terms	
Submit	Reset

Enrollment request submitted successfully!

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#)

Student Enrollment Form

Student Name

UK

Name must be at least 3 characters

Email

Email is required

Course ID

Preferred Semester

☐ I Agree to the Terms

Submit

Reset

HANDS-ON 5 – Reactive Forms — FormBuilder, FormGroup, FormArray & Custom

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Reactive Enrollment Form

Student Name

Utkarsh

Email

test@gmail.com

Email already taken.

Course ID

XX102

Course code starting with XX is not allowed.

Preferred Semester

Odd

Additional Courses

Add Another Course

Remove

☒ I Agree to the Terms

Submit

Elements

Console

81

Filter

Custom levels

81 Issues

81

3 messages

3 user mes...

No errors

No warnings

1 info

2 verbose

[vite] connecting... client:733

Angular debug_node-chunk.mjs:11068

r is running in development mode.

[vite] connected. client:827

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Reactive Enrollment Form

Student Name

Email

Course ID

Preferred Semester

☒ I Agree to the Terms

HANDS-ON 6 – Services & Dependency Injection

The screenshot displays a web browser at localhost:4200 showing the 'Student Course Portal' application. The page features a navigation bar with links: [Home](#), [Courses](#), [Profile](#), [Enroll](#), and [Reactive Enroll](#). The main heading is 'Student Course Portal'. Below it, there is an 'Enroll Now' button and a search bar labeled 'Search courses'. The page content includes a search status 'Searching for:', a description 'This portal helps students browse courses, manage enrollment, and view their academic profile.', and statistics: 'Courses Available: 5', 'Enrolled: 3', and 'GPA: 3.8'. A 'Course Summary' section shows 'Total Courses: 5'. A 'Notification' section displays 'Welcome to the Student Portal!'. The browser's developer console is open on the right, showing 14 issues. The console log includes messages from Vite, Angular, and the application's HomeComponent, such as '[vite] connecting...', 'Angular debug_node-chunk.mjs:11068 r is running in development mode.', '[vite] connected.', 'HomeComponent initialised - courses loaded', 'HomeComponent destroyed', and 'HomeComponent initialised - courses loaded'.

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Angular ID: 1 Code: ANG101 Credits: 4 Credits Passed UnenrollShow Details
Java ID: 2 Code: JAVA201 Credits: 3 Credits Failed EnrollShow Details

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Student Profile

Enrolled Courses

- Angular (ANG101)

HANDS-ON 7 – Angular Routing — Guards, Lazy Loading & Route Data

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Student Enrollment Form

Student Name

Email

Course ID

Preferred Semester Select

☐ I Agree to the Terms

Elements Console Sources Network

Filter

20,000 ms 40,000 ms 60,000 ms 80,000 ms 100,000 ms 120,000 ms 140,000 ms

Name	Status	Type	Initiator	Size	Time
reactive	200	docum...	Other	3.7 kB	26 ms
client	200	script	reactive:2	178 kB	6 ms
styleshe...	200	script	reactive:9	0.3 kB	7 ms
main.js	200	script	reactive:31	125 kB	9 ms
chunk-SQIA2N4W.js	200	script	reactive:32	3.1 kB	8 ms
chunk-4UEOPAWG.js	200	script	reactive:33	27.4 kB	11 ms
env.mjs	200	script	client:1	3.8 kB	2 ms
chunk-SG2LYVAH.js	200	script	main.js:4	1.1 kB	2 ms
@angular_platform-browser.js...	200	script	main.ts:1	9.8 kB	3 ms
@angular_core.js?v=8a33e398	200	script	app.config.ts:1	103 kB	7 ms
@angular_router.js?v=8a33e398	200	script	app.config.ts:2	217 kB	13 ms
@angular_forms.js?v=8a33e398	200	script	home.ts:2	152 kB	15 ms
@angular_common.js?v=8a33...	200	script	course-list.ts:2	18.8 kB	15 ms
?token=kO3svL8pup5g	101	websoc...	client:745	0.0 kB	Pending
chunk-XRRPNBHN.js?v=8a33e...	200	script	@angular_platform	140 kB	9 ms
chunk-GT36345P.js?v=8a33e398	200	script	@angular_platform	170 kB	10 ms
chunk-MZPJPS5C.js?v=8a33e398	200	script	@angular_platform	1,053 kB	18 ms
component?c=src%2Fapp%2Fc...	200	script	course-summary-	0.2 kB	2 ms
component?c=src%2Fapp%2Fc...	200	script	notification.ts:15	0.2 kB	8 ms
component?c=src%2Fapp%2Fc...	200	script	home.ts:19	0.2 kB	32 ms
component?c=src%2Fapp%2Fc...	200	script	course-card.ts:14	0.2 kB	32 ms
component?c=src%2Fapp%2Fc...	200	script	course-list.ts:17	0.2 kB	32 ms
component?c=src%2Fapp%2Fc...	200	script	course-detail.ts:15	0.2 kB	32 ms
component?c=src%2Fapp%2Fc...	200	script	student-profile.ts:	0.2 kB	32 ms
component?c=src%2Fapp%2Fc...	200	script	not-found.ts:9	0.2 kB	31 ms
component?c=src%2Fapp%2Fc...	200	script	courses-layout.ts:	0.2 kB	30 ms
component?c=src%2Fapp%2Fc...	200	script	header.ts:11	9.2 kB	32 ms
component?c=src%2Fapp%2Fa...	200	script	app.ts:11	0.2 kB	31 ms
component?c=src%2Fapp%2Fc...	200	script	reactive-enrollme	0.2 kB	17 ms
chunk-UBFML4NS.js	200	script	enrollment.routes	31.7 kB	3 ms
component?c=src%2Fapp%2Fc...	200	script	enrollment-form.t	0.2 kB	2 ms

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Courses

Search Course

Angular

ID: 1

Code: ANG101

Credits: 4 Credits

Passed



Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Student Course Portal

Enroll Now

Search courses

Searching for:

This portal helps students browse courses, manage enrollment, and view their academic profile.

Courses Available: 5

Enrolled: 3

GPA: 3.8

Course Summary

Total Courses: 5

Notification

Welcome to the Student Portal!



Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Courses

Course Details

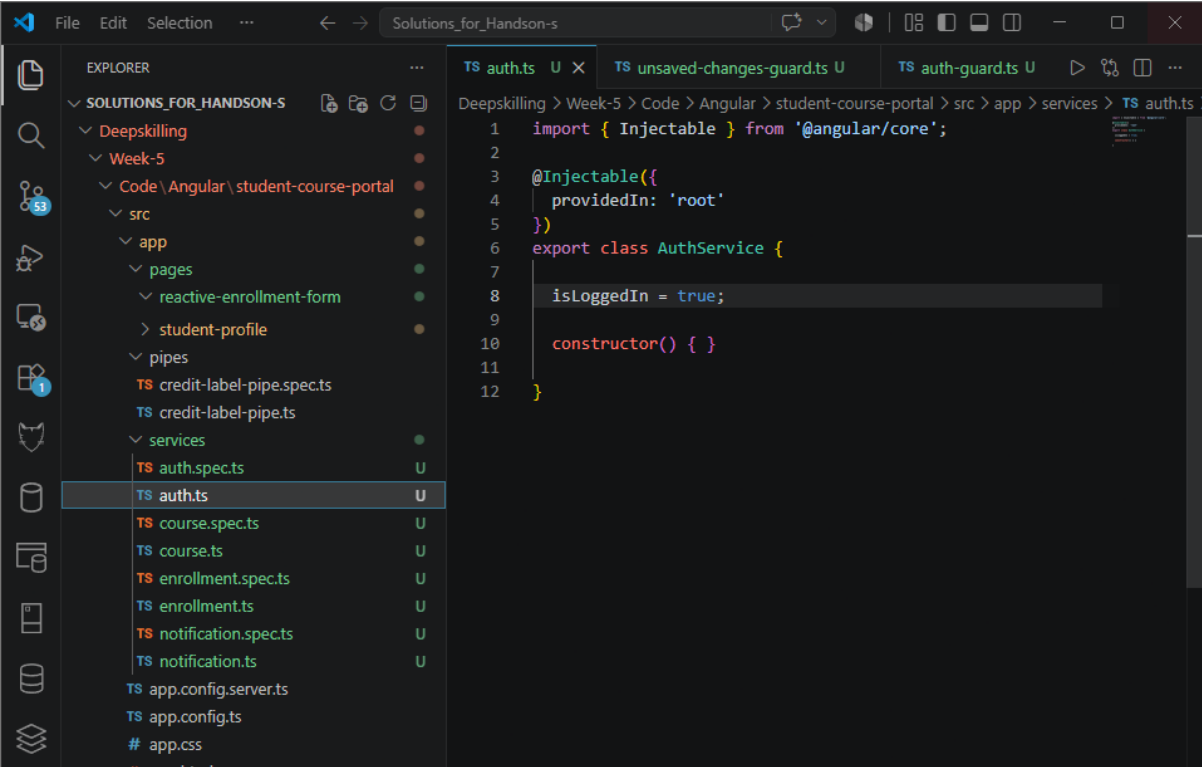
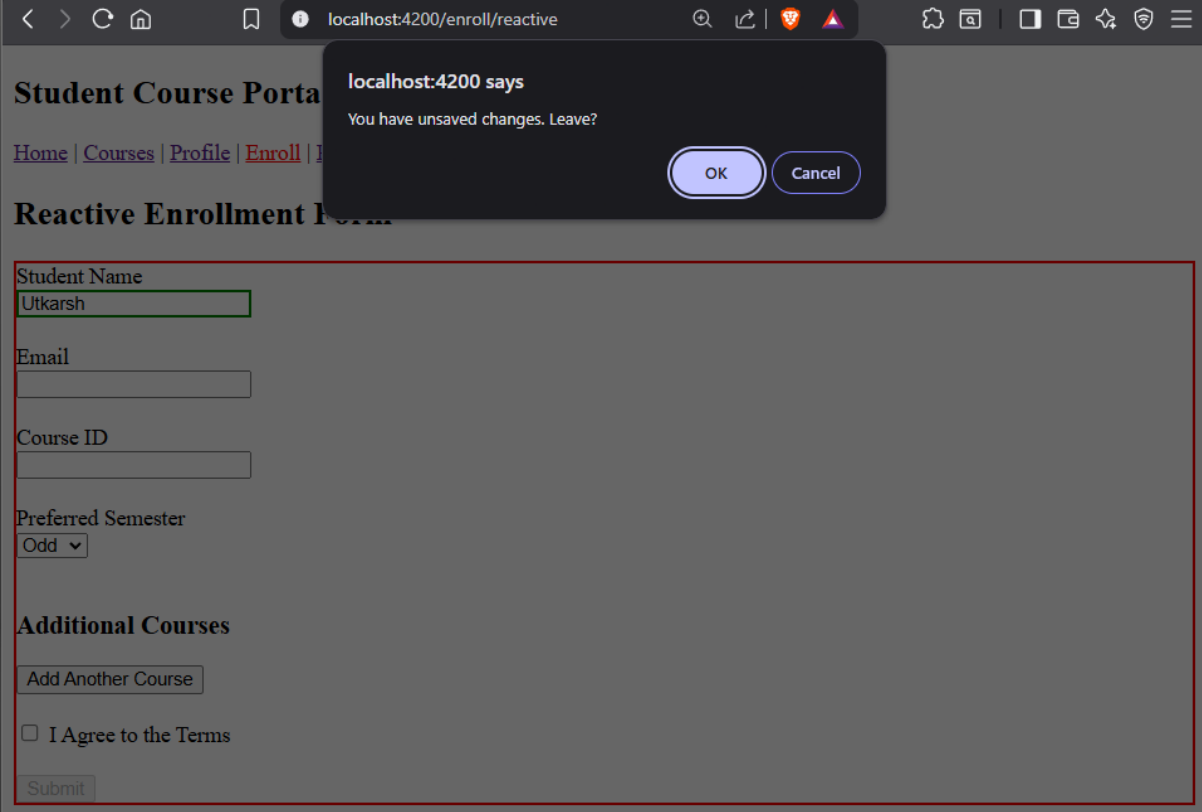
ID: 1

Name: Angular

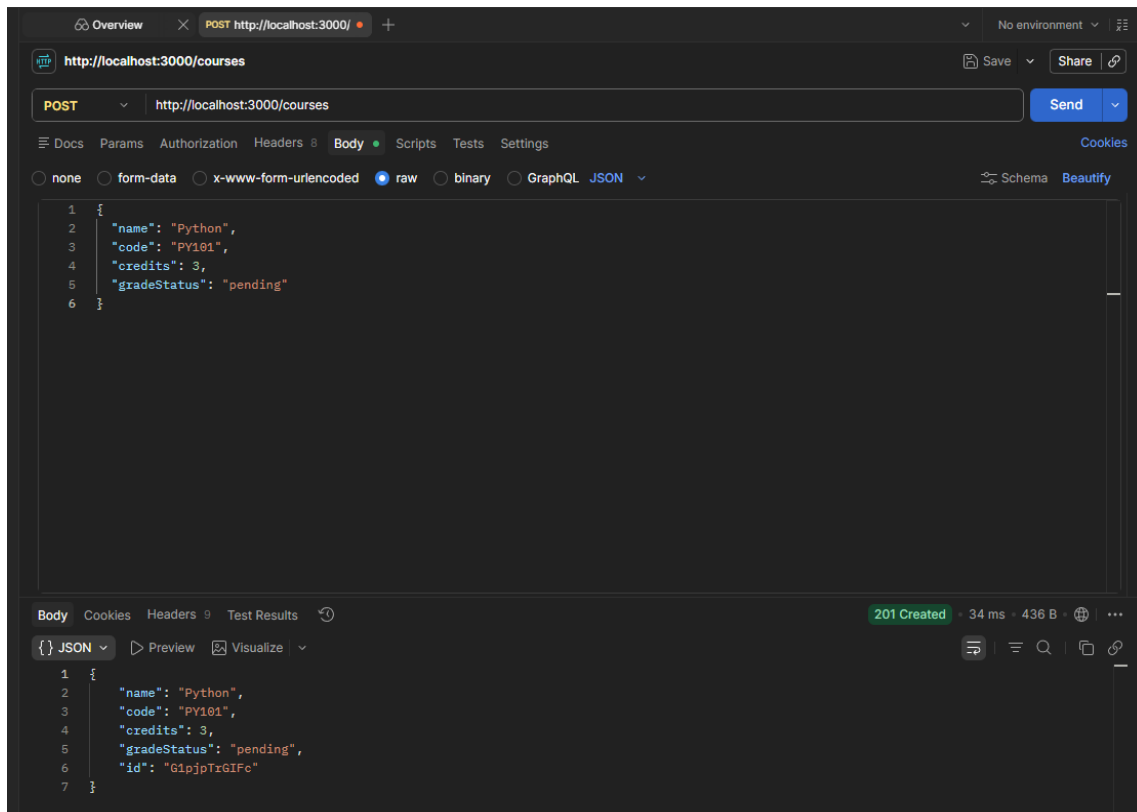
Code: ANG101

Credits: 4

Status: passed



HANDS-ON 8 -- HTTP Client — API Integration, Observables & Interceptors



Overview POST http://localhost:3000/ No environment

http://localhost:3000/courses Save Share

POST Send

Docs Params Authorization Headers 8 **Body** Scripts Tests Settings Cookies

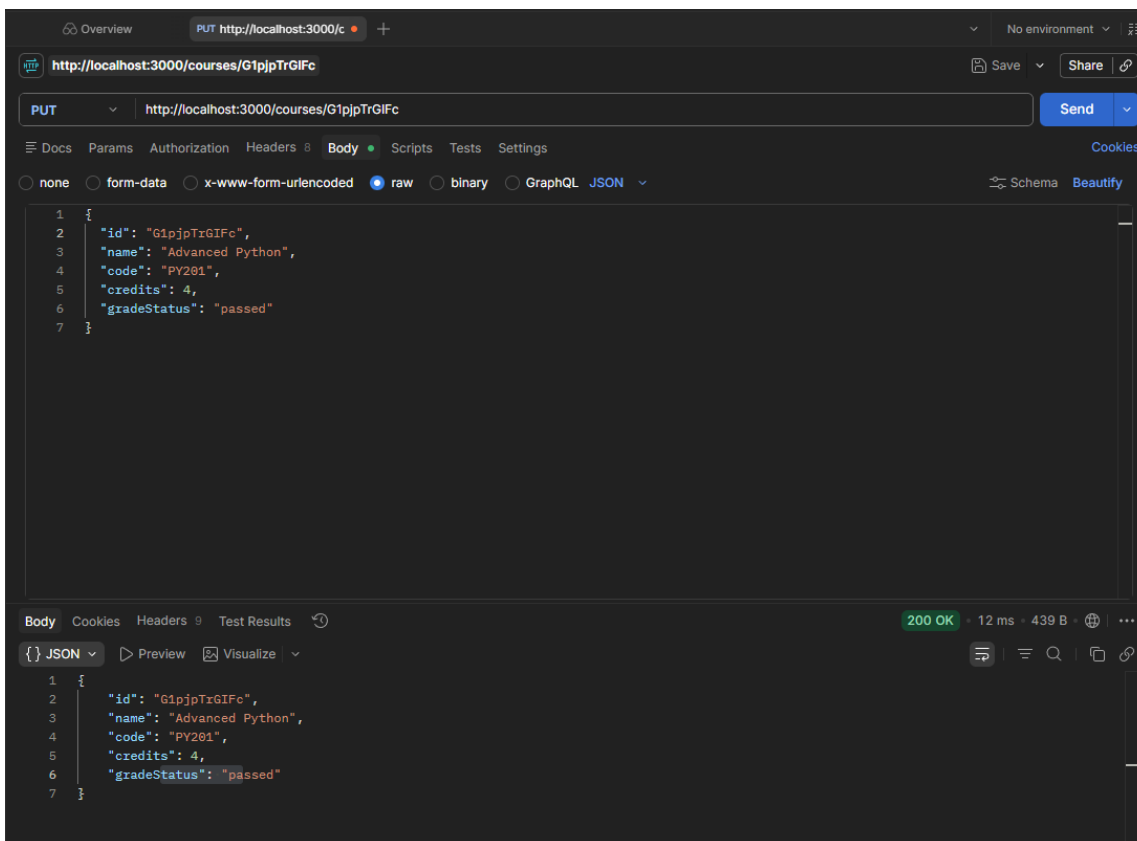
☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL ☐ JSON Schema Beautify

```
1 {
2   "name": "Python",
3   "code": "PY101",
4   "credits": 3,
5   "gradeStatus": "pending"
6 }
```

Body Cookies Headers 9 Test Results 201 Created · 34 ms · 436 B

JSON Preview Visualize

```
1 {
2   "name": "Python",
3   "code": "PY101",
4   "credits": 3,
5   "gradeStatus": "pending",
6   "id": "G1pjpTrGIFc"
7 }
```



Overview PUT http://localhost:3000/c No environment

http://localhost:3000/courses/G1pjpTrGIFc Save Share

PUT Send

Docs Params Authorization Headers 8 **Body** Scripts Tests Settings Cookies

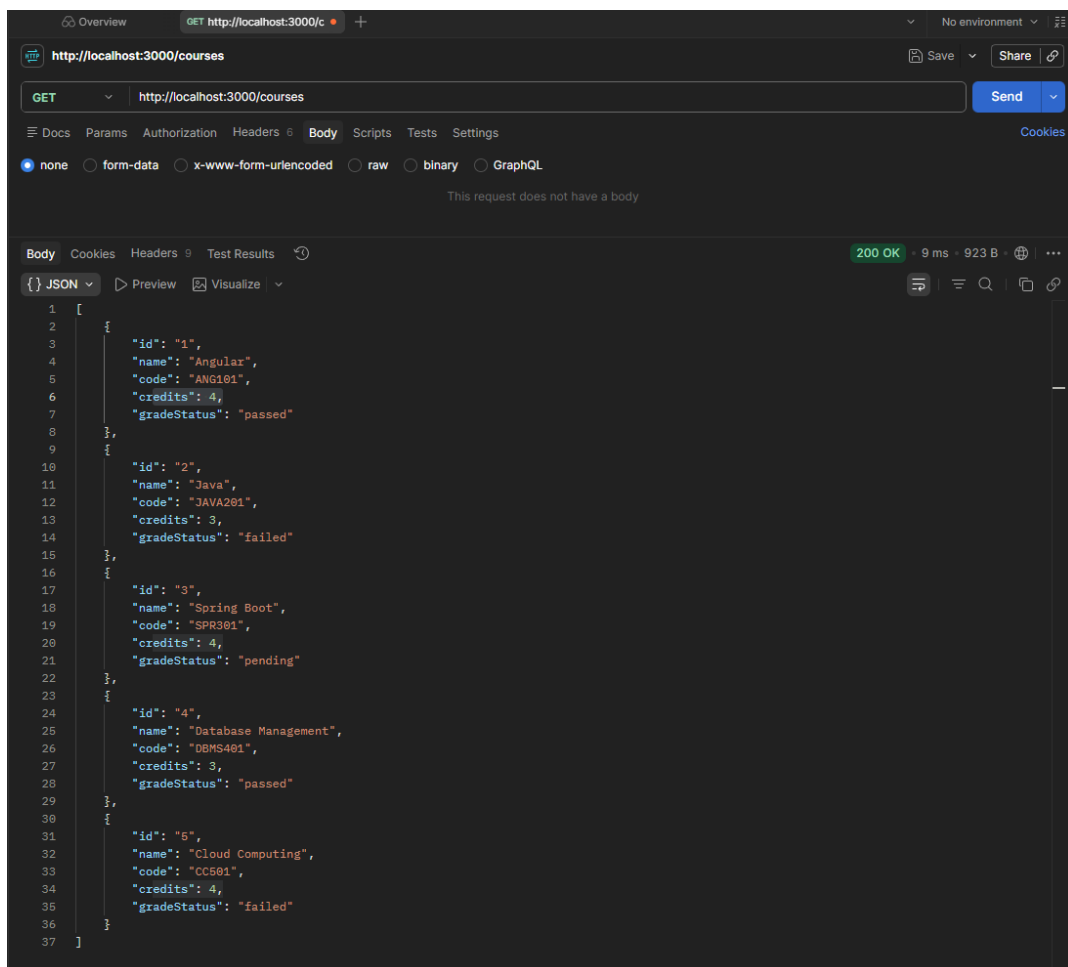
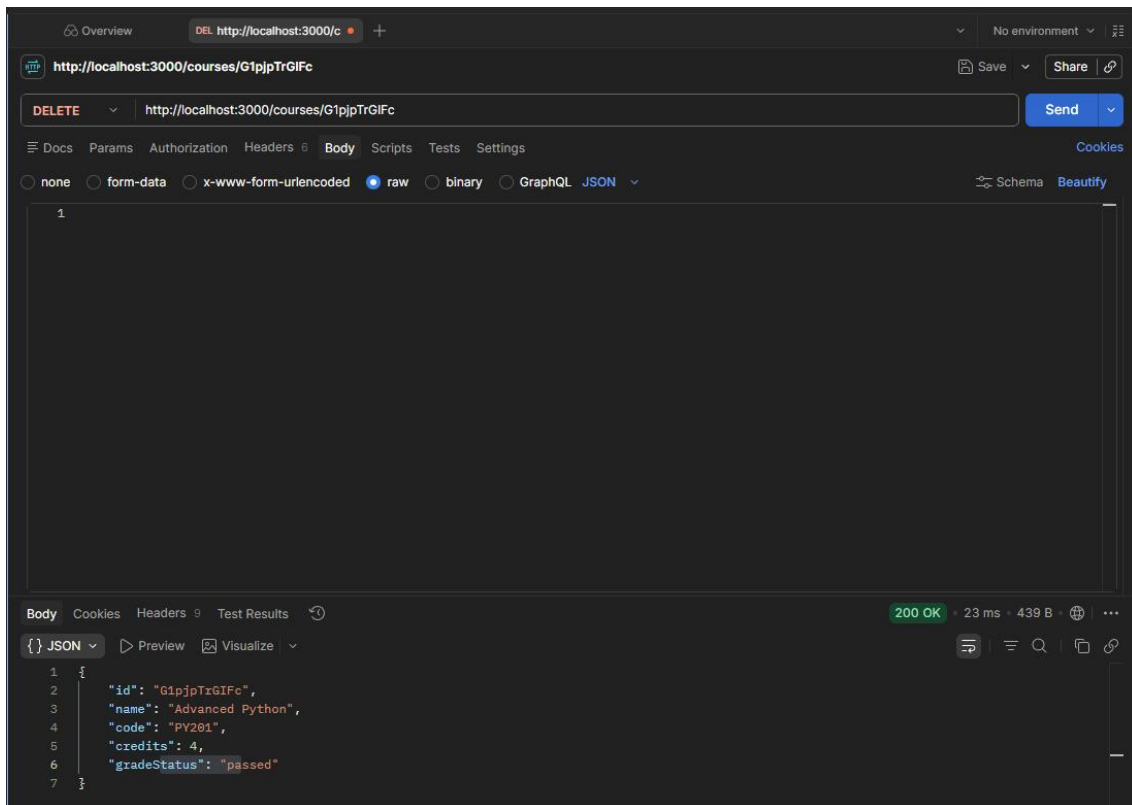
☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL ☐ JSON Schema Beautify

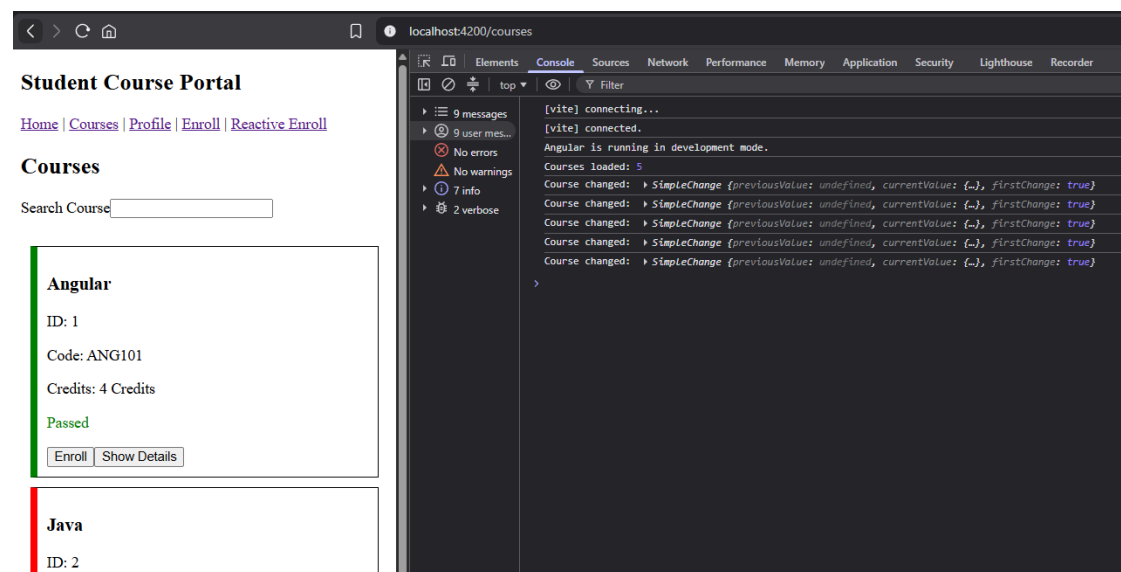
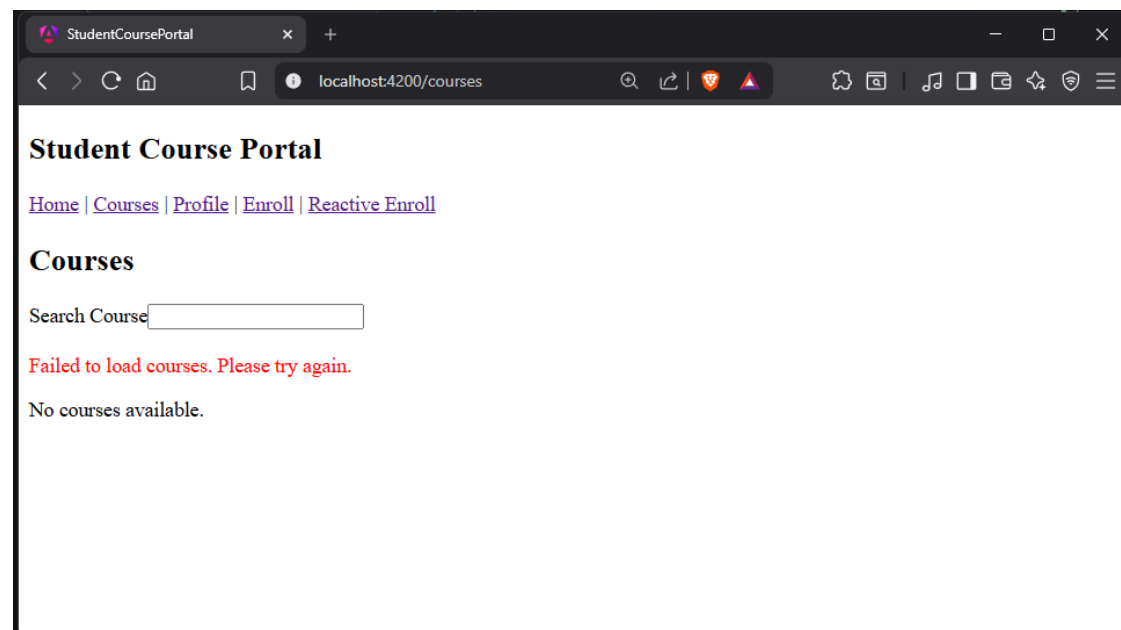
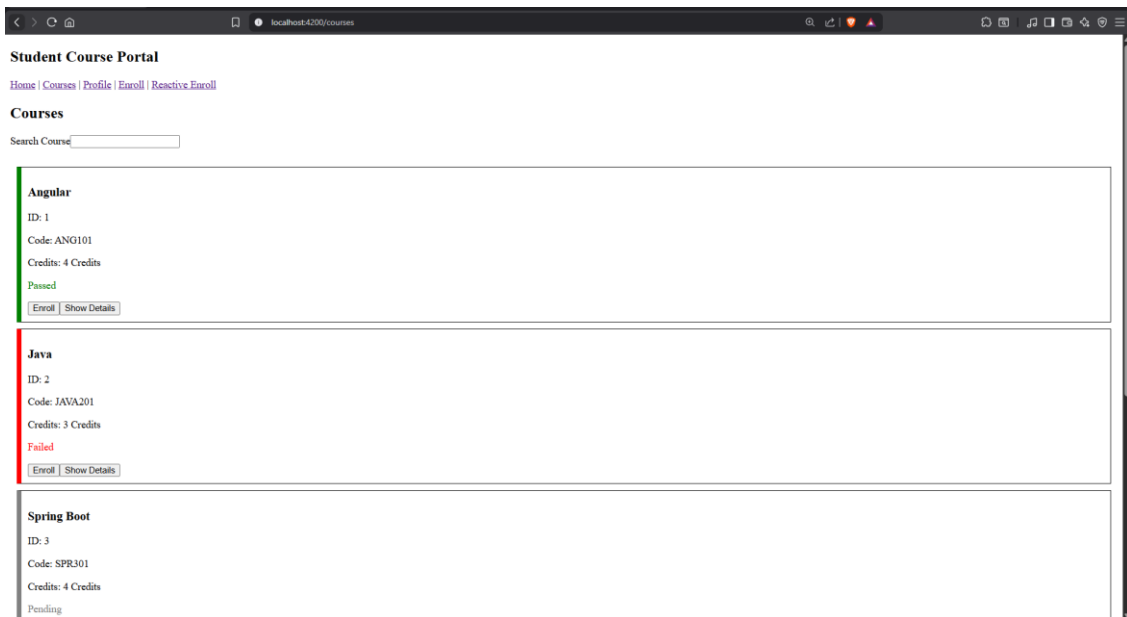
```
1 {
2   "id": "G1pjpTrGIFc",
3   "name": "Advanced Python",
4   "code": "PY201",
5   "credits": 4,
6   "gradeStatus": "passed"
7 }
```

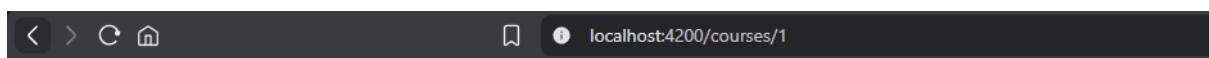
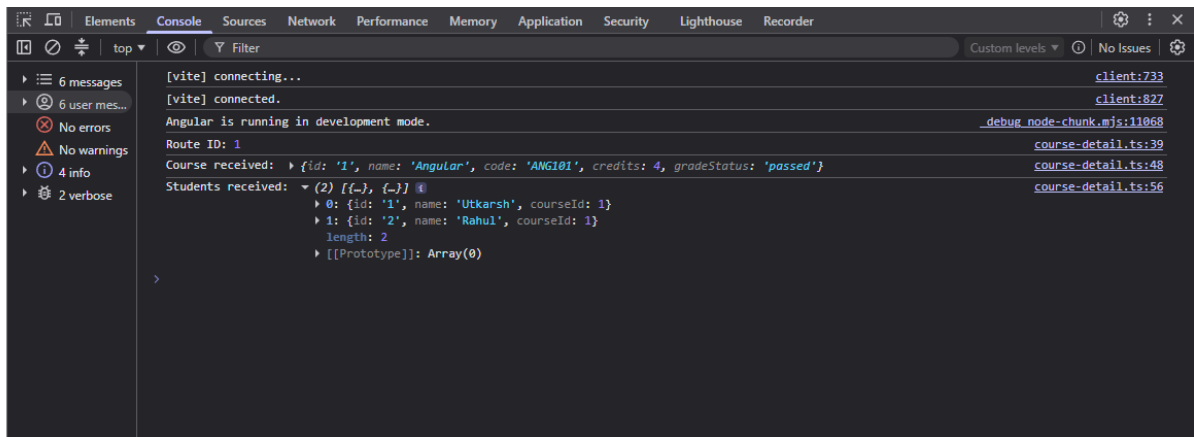
Body Cookies Headers 9 Test Results 200 OK · 12 ms · 439 B

JSON Preview Visualize

```
1 {
2   "id": "G1pjpTrGIFc",
3   "name": "Advanced Python",
4   "code": "PY201",
5   "credits": 4,
6   "gradeStatus": "passed"
7 }
```







Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Courses

Course Details

ID: 1

Name: Angular

Code: ANG101

Credits: 4

Status: passed

Enrolled Students

- Utkarsh
- Rahul

▼ Request Headers	☐ Raw
Accept	application/json, text/plain, */*
Accept-Encoding	gzip, deflate, br, zstd
Accept-Language	en-US,en;q=0.9
Authorization	Bearer mock-token-12345
Cache-Control	no-cache
Connection	keep-alive
Host	localhost:3000
Origin	http://localhost:4200
Pragma	no-cache
Referer	http://localhost:4200/
Sec-Ch-Ua	"Not;A=Brand";v="8", "Chromium";v="150", "Brave";v="150"
Sec-Ch-Ua-Mobile	?0
Sec-Ch-Ua-Platform	"Windows"
Sec-Fetch-Dest	empty
Sec-Fetch-Mode	cors
Sec-Fetch-Site	same-site
Sec-Gpc	1
User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/150.0.0.0 Safari/537.36

[<](#) [>](#) [↺](#) [🏠](#) [🔖](#) [📘 localhost:4200/courses](#)

Student Course Portal

[Home](#) | [Courses](#) | [Profile](#) | [Enroll](#) | [Reactive Enroll](#)

Loading...

Courses

Search Course

Loading courses...

No courses available.

Student Course Portal

Home | Courses | Profile | Enroll | Reactive Enroll

Courses

Course Details

ID: 1

Name: Angular

Code: ANG101

Credits: 4

Status: passed

Enrolled Students

- Utkarsh
- Rahul

Name	Status	Type	Initiator	Size	Time
@angular.platform-browser.js?v=8a2496dc	200	script	main.js:13	14.2 kB	7 ms
@angular.common.js?v=8a2496dc	200	script	main.js:13	102 kB	22 ms
@angular_router.js?v=8a2496dc	200	script	app.config.js:2	218 kB	25 ms
@angular_common_http.js?v=8a2496dc	200	script	app.config.js:3	12.6 kB	23 ms
@angular_forms.js?v=8a2496dc	200	script	home.js:2	152 kB	22 ms
@angular_common.js?v=8a2496dc	200	script	notification.js:2	22.4 kB	23 ms
ng.js?v=8a2496dc	200	script	notification.js:2	23.1 kB	24 ms
ng_router.js?v=8a2496dc	200	script	course-detail.js:4	15.8 kB	21 ms
.*?Token=Wj-9PQk1g1uh	101	websocket	client.js:45	0.0 kB	Pending
chunk-107MOSD.js?v=8a2496dc	200	script	@angular.platform-browser.js:37	55.3 kB	9 ms
chunk-AS8T7N0V.js?v=8a2496dc	200	script	@angular.platform-browser.js:38	88.8 kB	19 ms
chunk-C0MB0TDE.js?v=8a2496dc	200	script	@angular.platform-browser.js:39	147 kB	18 ms
chunk-PJ2Z582.js?v=8a2496dc	200	script	@angular.platform-browser.js:42	4.5 kB	15 ms
chunk-C26461NV.js?v=8a2496dc	200	script	@angular.platform-browser.js:43	96.1 kB	23 ms
chunk-4VCCDQOQ.js?v=8a2496dc	200	script	@angular.platform-browser.js:44	1.2 kB	20 ms
chunk-H8EEVGT.js?v=8a2496dc	200	script	@angular.platform-browser.js:45	28.1 kB	22 ms
chunk-UTEDC2PH.js?v=8a2496dc	200	script	@angular.platform-browser.js:46	148 kB	28 ms
component1-c-app%2fcomponents%2fcourse-summary-widg.js?v=8a2496dc	200	script	course-summary-widg.js:11	0.2 kB	6 ms
component1-c-app%2fcomponents%2fnotification.js?v=8a2496dc	200	script	notification.js:3	0.2 kB	6 ms
component1-c-app%2fcomponents%2fhome.js?v=8a2496dc	200	script	home.js:19	0.2 kB	5 ms
component1-c-app%2fcomponents%2fcourse-card.js?v=8a2496dc	200	script	course-card.js:14	0.2 kB	5 ms
component1-c-app%2fcomponents%2fcourse-list.js?v=8a2496dc	200	script	course-list.js:22	0.2 kB	5 ms
component1-c-app%2fcomponents%2fcourse-detail.js?v=8a2496dc	200	script	course-detail.js:18	0.2 kB	4 ms
component1-c-app%2fcomponents%2fstudent-profile.js?v=8a2496dc	200	script	student-profile.js:14	0.2 kB	5 ms
component1-c-app%2fcomponents%2fnot-found.js?v=8a2496dc	200	script	not-found.js:9	0.2 kB	2 ms
component1-c-app%2fcomponents%2fcourses-layout.js?v=8a2496dc	200	script	courses-layout.js:10	0.2 kB	3 ms
component1-c-app%2fcomponents%2fheader.js?v=8a2496dc	200	script	header.js:11	0.2 kB	3 ms
component1-c-app%2fcomponents%2fapp.js?v=8a2496dc	200	script	app.js:13	9.7 kB	3 ms
1	204	preflight	Preflight @	0.0 kB	2 ms
1	200	fetch	course-detail.js:40	0.4 kB	4 ms
1	200	fetch	course-detail.js:40	0.0 kB	2 ms
1	204	preflight	Preflight @	0.0 kB	3 ms
1	200	fetch	course-detail.js:40	0.5 kB	2 ms
1	204	preflight	Preflight @	0.0 kB	2 ms
1	200	fetch	course-detail.js:40	0.5 kB	2 ms

44 requests | 2.4 MB transferred | 2.4 MB resources | Fetches: 365 ms | DOMContentLoaded: 303 ms | Load: 308 ms

Student Course Portal

Home | Courses | Profile | Enroll | Reactive Enroll

Courses

Search Course

Failed to load courses. Please try again.

No courses available.

9 messages

6 user messages

6 errors

No warnings

1 info

2 verbose

[vite] connecting... client:733

[vite] connected. client:827

Angular debug node-chunk.mjs:11068

lar is running in development mode.

ERROR RuntimeError: main.ts:5

NG0100: ExpressionChangedAfterItHasBeenCheckedError: Expression has changed after it was checked. Previous value: 'false'. Current value: 'true'. Expression location: _App component. Find more at https://v21.angular.dev/errors/NG0100 at App_Template (app.html:3:7)

Show ignore-listed frames

ERROR _effect-chunk2.mjs:2601

RuntimeError: NG0100: ExpressionChangedAfterItHasBeenCheckedError: Expression has changed after it was checked. Previous value: 'false'. Current value: 'true'. Expression location: _App component. Find more at https://v21.angular.dev/errors/NG0100 at App_Template (app.ts:12:15)

Show ignore-listed frames

course.ts:28

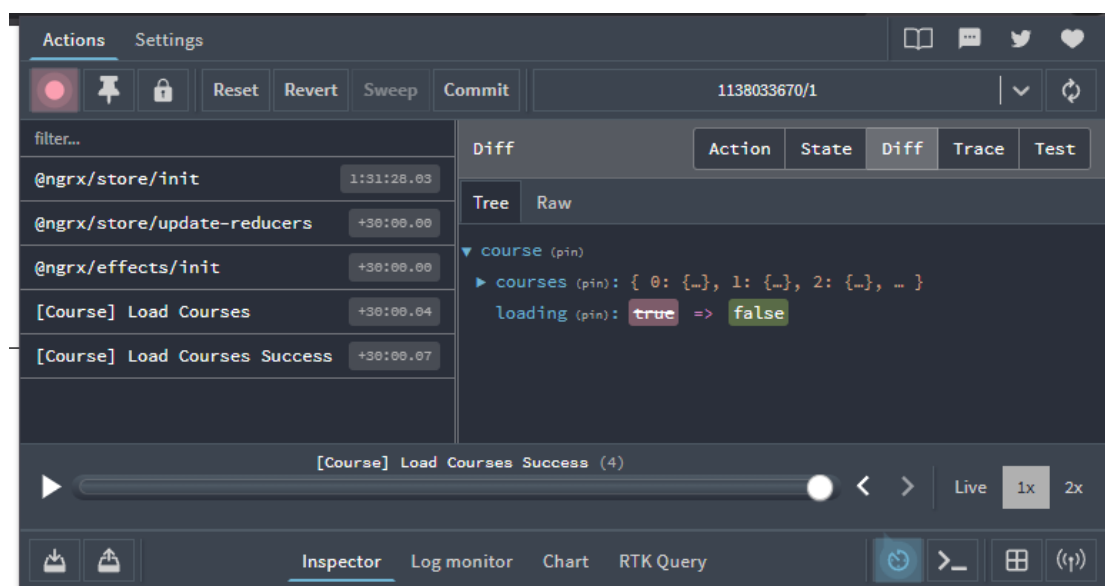
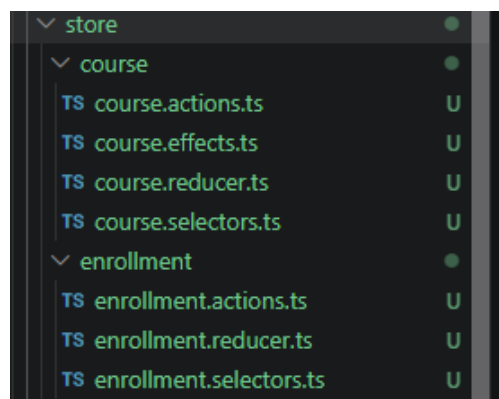
HttpErrorResponse {headers: Headers, status: 0, statusText: 'Unknown Error', url: 'http://localhost:3000/courses', ok: false, ...}

HANDS-ON 9 -- State Management — NgRx Store, Actions, Reducers, Effects & Selectors

```
ollment.selectors.ts U  courses-layout.html M  TS main.ts  TS courses-layout.ts  TS app.config.ts M X  TS course.effects.ts U  TS course.selectors.ts U

Deepskilling > Week-5 > Code > Angular > student-course-portal > src > app > TS app.config.ts > ...
1  import { ApplicationConfig, provideBrowserGlobalErrorListeners } from '@angular/core';
2  import { provideRouter } from '@angular/router';
3  import { provideHttpClient, withFetch, withInterceptors } from '@angular/common/http';
4  import { routes } from './app.routes';
5  import { authInterceptor } from './interceptors/auth-interceptor';
6  import { loadingInterceptor } from './interceptors/loading-interceptor';
7  import { errorInterceptor } from './interceptors/error-interceptor';
8  import { provideStore } from '@ngrx/store';
9  import { provideState } from '@ngrx/store';
10 import { provideEffects } from '@ngrx/effects';
11 import { courseReducer } from './store/course/course.reducer';
12 import { CourseEffects } from './store/course/course.effects';
13 import { provideStoreDevtools } from '@ngrx/store-devtools';
14 import { enrollmentReducer } from './store/enrollment/enrollment.reducer';
15
16 export const appConfig: ApplicationConfig = {
17   providers: [
18     provideBrowserGlobalErrorListeners(),
19     provideRouter(routes),
20     provideStore(),
21     provideStoreDevtools({
22       maxAge: 25}),
23     provideEffects([CourseEffects]),
24     provideState('course', courseReducer),
25     provideState('enrollment', enrollmentReducer),
26     provideHttpClient(
27       withFetch(),
28       withInterceptors([authInterceptor, loadingInterceptor, errorInterceptor])
29     )
30   ]
31 , };

```



Actions Settings

1138033670/1

filter...

@ngrx/store/init 1:34:52.82

@ngrx/store/update-reducers +30:00.00

@ngrx/effects/init +30:00.00

State

Action State Diff Trace Test

Tree Chart Raw

```
1 {
2   course: {
3     courses: [],
4     loading: false,
5     error: null
6   },
7   enrollment: {
8     enrolledCourseIds: []
9   }
10 }
```

@ngrx/effects/init (2)

Live 1x 2x

Inspector Log monitor Chart RTK Query

StudentCourseFrontal

localhost:4200/courses

Angular

ID: 1

Code: ANG101

Credits: 4 Credits

Passed

Unenroll Show Details

Java

ID: 2

Code: JAVA201

Credits: 3 Credits

Failed

Enroll Show Details

Course Data

Actions Settings

1138033750/1

filter...

@ngrx/store/init 0:14:130.42

@ngrx/store/update-reducers +30:00.00

@ngrx/effects/init +30:00.00

[Course] Load Courses +30:00.00

[Course] Load Courses Success +30:00.02

[Enrollment] Enroll In Course +30:00.08

State

Action State Diff Trace Test

Tree Chart Raw

```
course {unit}
  courses {unit} [[-], [-], [-], [-], [-]]
  loading {unit} false
  error {unit} null
  enrollment {unit}
    enrolledCourseIds {unit} ["1"]
```

[Enrollment] Enroll In Course (5)

Live 1x 2x

Inspector Log monitor Chart RTK Query

HANDS-ON 10 -- Unit Testing Angular Applications — Jasmine, Karma & TestBed

```
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-5\Code\Angular\student-course-portal> ng test

Initial chunk files
spec-app-pages-enrollment-form-enrollment-form.js 17.70 kB
spec-app-pages-reactive-enrollment-form-reactive-enrollment-form.js 11.78 kB
chunk-F5M2XEQ.js - 11.36 kB
spec-app-pages-course-list-course-list.js 9.86 kB
spec-app-pages-course-detail-course-detail.js 7.40 kB
spec-app-pages-home-home.js 5.81 kB
spec-app-pages-student-profile-student-profile.js 4.30 kB
chunk-CG2BV68D.js - 4.07 kB
spec-app-app.js 4.01 kB
chunk-ZACMF7.js - 2.40 kB
chunk-4YTDAS3D.js - 2.01 kB
chunk-BPHALSNT.js - 2.00 kB
spec-app-pages-courses-layout-courses-layout.js 1.99 kB
chunk-R7PXWPPX.js - 1.91 kB
spec-app-pages-not-found-not-found.js 1.85 kB
chunk-KQLELTDH.js - 1.73 kB
init-testbed.js 1.21 kB
chunk-R7HQHMQ.js 992 bytes
vitest-mock-patch.js 974 bytes
chunk-X45QZX36.js - 903 bytes
spec-app-components-course-card-course-card.js 889 bytes
spec-app-guards-auth-guard.js 871 bytes
spec-app-guards-unsaved-changes-guard.js 842 bytes
spec-app-components-course-summary-widget-course-summary-widget.js 761 bytes
chunk-V0OG7V4Z.js - 743 bytes
spec-app-components-header-header.js 719 bytes
spec-app-components-notification-notification.js 697 bytes
spec-app-services-enrollment.js 630 bytes
spec-app-services-notification.js 492 bytes
spec-app-services-course.js 462 bytes
spec-app-services-auth.js 452 bytes
spec-app-pipes-credit-label-pipe.js 344 bytes
styles.css 56 bytes

Initial total 102.23 kB

Application bundle generation complete. [2.609 seconds] - 2026-07-24T14:35:44.946Z

Watch mode enabled. Watching for file changes...

DEV v4.1.10 D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-5\Code\Angular\student-course-portal

> student-course-portal src/app/components/course-card/course-card.spec.ts [queued]
> student-course-portal src/app/components/course-summary-widget/course-summary-widget.spec.ts [queued]
```

```
PS D:\Cogniz_folder\Solutions_for_Handson-s\Deepskilling\Week-5\Code\Angular\student-course-portal> ng test

✓ student-course-portal src/app/components/course-summary-widget/course-summary-widget.spec.ts (1 test) 114ms
✓ student-course-portal src/app/components/header/header.spec.ts (1 test) 100ms
✓ student-course-portal src/app/components/course-card/course-card.spec.ts (1 test) 134ms
✓ student-course-portal src/app/pages/course-detail/course-detail.spec.ts (1 test) 82ms
✓ student-course-portal src/app/pages/courses-layout/courses-layout.spec.ts (1 test) 11ms
✓ student-course-portal src/app/pages/reactive-enrollment-form/reactive-enrollment-form.spec.ts (1 test) 160ms

> student-course-portal src/app/app.spec.ts [queued]
> student-course-portal src/app/components/notification/notification.spec.ts 1/1
> student-course-portal src/app/guards/unsaved-changes-guard.spec.ts [queued]
> student-course-portal src/app/pages/course-list/course-list.spec.ts (1 test) 15ms
✓ student-course-portal src/app/pages/student-profile/student-profile.spec.ts (1 test) 14ms
✓ student-course-portal src/app/pages/not-found/not-found.spec.ts (1 test) 18ms
✓ student-course-portal src/app/guards/unsaved-changes-guard.spec.ts (1 test) 4ms
stdout | src/app/pages/home/home.spec.ts > Home > should create
HomeComponent initialised – courses loaded

stdout | src/app/pages/home/home.spec.ts > Home > should create
Courses loaded: 5

✓ student-course-portal src/app/pages/enrollment-form/enrollment-form.spec.ts (1 test) 163ms
✓ student-course-portal src/app/services/auth.spec.ts (1 test) 4ms
stdout | src/app/pages/home/home.spec.ts > Home > should create
Courses loaded: 5

✓ student-course-portal src/app/services/notification.spec.ts (1 test) 4ms
stdout | src/app/pages/home/home.spec.ts > Home > should create
HomeComponent destroyed

✓ student-course-portal src/app/services/course.spec.ts (1 test) 6ms
✓ student-course-portal src/app/services/enrollment.spec.ts (1 test) 6ms
✓ student-course-portal src/app/pages/home/home.spec.ts (1 test) 153ms
✓ student-course-portal src/app/guards/auth-guard.spec.ts (1 test) 1ms
✓ student-course-portal src/app/pipes/credit-label-pipe.spec.ts (1 test) 1ms
✓ student-course-portal src/app/app.spec.ts (1 test) 37ms
✓ student-course-portal src/app/pages/course-list/course-list.spec.ts (1 test) 97ms

Test Files 20 passed (20)
Tests 20 passed (20)
Start at 20:05:45
Duration 5.08s (transform 872ms, setup 9.04s, import 2.06s, tests 1.12s, environment 60.83s)

PASS Waiting for file changes...
press h to show help, press q to quit
```

